Physics Formula Sheet — TEST BENCH

Sample layout used only to test preview, video and listing tools

1. Kinematics

$$v = v_0 + at$$
$$x = x_0 + v_0 t + \tfrac{1}{2} a t^2$$
$$v^2 = v_0^2 + 2a(x - x_0)$$

2. Dynamics

$$F = ma \qquad W = mg$$
$$f_k = \mu_k N \qquad p = mv$$

3. Energy

$$K = \tfrac{1}{2} m v^2 \qquad U = mgh$$
$$W = F d \cos\theta \qquad P = \tfrac{W}{t}$$

4. Circular motion

$$a_c = \tfrac{v^2}{r} \qquad F_c = m\tfrac{v^2}{r}$$
$$\omega = 2\pi f \qquad T = \tfrac{1}{f}$$

5. Fluids

$$P = \tfrac{F}{A} \qquad P = P_0 + \rho g h$$
$$A_1 v_1 = A_2 v_2$$

6. Waves

$$v = f\lambda \qquad I \propto A^2$$

Worked example (illustrative)

A cart starts from rest with constant acceleration a. After time t its speed is $v = at$ and the distance covered is $x = \tfrac{1}{2} a t^2$. (Symbols only: this test sheet carries no numerical data.)

Common pitfalls

Mixing degrees and radians · forgetting the sign of g · adding vectors as scalars.

Electricity & Optics — TEST BENCH

7. Electrostatics

$F = k\frac{q_1 q_2}{r^2}$
$E = \frac{F}{q}$ $V = k\frac{q}{r}$

8. Circuits

$V = IR$ $P = IV$
Series: $R = R_1 + R_2$
Parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

9. Optics

$n = \frac{c}{v}$
$n_1 \sin\theta_1 = n_2 \sin\theta_2$
$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

10. Units

Quantity	SI unit
Force	newton (N)
Energy	joule (J)
Power	watt (W)
Charge	coulomb (C)

Test bench only. Not a product, not for sale.